



HESHANI WIJESINGHE

DATA SCIENCE GRADUATE

+94 762328652 | heshani03123@gmail.com | github.com/Heshi4 | linkedin.com/in/heshaniwijesinghe

SUMMARY

Data Science graduate with hands-on experience in machine learning, data engineering, and process automation. Proven ability to develop scalable solutions that reduce manual effort and improve operational efficiency — including an 83% reduction in mapping time through a Selenium-based automation bot recognized with a “Best Solution” award at Tech Wave 2025. Skilled in building end-to-end data solutions, from data preprocessing and modeling to visualization and deployment.

EDUCATION

Bachelor of Science in Information Technology Specializing in Data Science 2022 – 2026
Sri Lanka Institute of Information Technology (SLIIT)

GCE Advanced & Ordinary Level, Pearson Edexcel London Examinations Completed 2021
Gateway College - IGCSE & GCSE

WORK EXPERIENCE

Intern - Work Study MCI Apr 2025 – Oct 2025
MAS Holdings

- Developed a Selenium-based Python automation bot integrated with APIs, reducing SMV mapping time from **30 days to 5 days (~83% reduction)** and significantly improving operational efficiency
- Built automation solutions using Python, Excel VBA, Power Automate, and API integrations, including an alert system for FI date anomalies to improve accuracy and reduce manual monitoring
- Led end-to-end development and deployment of automation workflows, collaborating with stakeholders to align solutions with business requirements
- Recognized for impact and innovation: **Best Solution** (Tech Wave 2025 – Smart Creation category) and **Silver Award** (Future Stars at MAS Intimates)

PROJECTS & RESEARCH

EDGE AI Based Online Examination Fraud Detection System Research Project

- Developed an Edge AI audio monitoring module for detecting suspicious exam behavior.
- Integrated ECAPA-TDNN for speaker verification using voice embeddings and cosine similarity.
- Applied YAMNet with transfer learning to extract audio features.
- Built a neural network classifier (TensorFlow/Keras) to classify audio as self-speaking, whispering, lecture, or noise.
- Optimized models using quantization and TorchScript for efficient edge deployment.
- Contributed to a real-time AI proctoring system combining audio, eye gaze, and head pose detection.

Technologies: Python, TensorFlow, PyTorch, Librosa, Edge AI, Deep Learning

Diabetes Prediction Model

3rd Year - 1st Semester

- Preprocessed and cleaned a Kaggle dataset by handling missing values, duplicates, and noise.
- Trained and compared KNN, Random Forest, Logistic Regression, and Neural Network models.
- Evaluated model performance using F1-score, precision, recall, and ROC-AUC; identified Random Forest as the best-performing model.
- Developed and deployed an interactive Streamlit application for real-time predictions.

Technologies: Python, Scikit-learn, TensorFlow, Streamlit

Healthcare Data Warehouse & BI System

3rd Year - 2nd Semester

- Collected, cleaned, and validated healthcare datasets for ETL processing.
- Designed and implemented ETL pipelines using SSIS, and built dimensional data models and OLAP cubes using SSAS.
- Developed interactive Power BI dashboards for patient insights and KPIs, enabling efficient, data-driven reporting.

Technologies: SQL Server, SSIS, SSAS, Power BI, Excel

Movie Recommendation System

3rd Year - 1st Semester

- Pre-processed two datasets, handling missing values and inconsistencies in Jupyter Notebook.
- Trained a machine learning model and exported it with Pickle for web integration.
- Developed a movie recommendation website with an HTML/CSS frontend and a Python backend.

Technologies: Pandas, Scikit-Learn, Python, HTML, CSS

Real Time Greenhouse Monitoring System With IOT & Visual Analytics

4th Year - 2nd Semester

- Developed an IoT system using ESP32, MQTT, and Firebase to monitor temperature, humidity, soil moisture, light, and air quality in real time across multiple greenhouse zones.
- Built interactive Power BI dashboards to visualize environmental trends with threshold-based alerts to detect zone-wise anomalies and support data-driven decisions.

Technologies: ESP32, MQTT, DHT22, LDR, MQ-135, Soil Moisture Sensor, Python, JavaScript, Power BI, Excel, Figma, Firebase

SKILLS

- Programming: Python, Java, C, C++
- Data Science & ML: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch
- Data Engineering: SQL Server, SSIS, SSAS
- Visualization: Power BI, Excel
- Automation: Selenium, Power Automate, Excel VBA
- Cloud: AWS, Azure

CERTIFICATIONS

- Career Essentials in Data Analysis – Microsoft & LinkedIn
- Career Essentials in Generative AI – Microsoft & LinkedIn
- Python for Beginners – University of Moratuwa
- Data Science Job Simulation – British Airways
- Trinity College London – Piano Examinations (Grades 1–7)